ASSIGNMENT 3 | QUIZ MASTER

Course	Web Development Laboratory
Experiment No.	Experiment – 3
Title	Quiz master website using HTML, CSS, & Flask
Year / Semester	B.Tech | Semester - IV
Department	Computer Science & Engineering
Faculty Incharge	Miss Mansi Kajal
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Enrollment No.	2401010201_____
Section / Batch	_Btech cse B_____

CODES;

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Quiz Master</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}"/>
</head>
<body>
  <div class="hero">
    <div class="hero-card">
      <div class="logo">□</div>
      <h1>Quiz Master</h1>
      <p class="subtitle">Test your knowledge with 5 random questions.<br>See how
well you score!</p>

      <div class="info-row">
        <div class="info-box"><span>📝</span> 5 Questions</div>
        <div class="info-box"><span>□</span> No Time Limit</div>
        <div class="info-box"><span>🎯</span> MCQ Format</div>
      </div>

      <a href="/quiz" class="btn-start">Start Quiz →</a>
    </div>
  </div>
</body>
```

```
</html>
```

Quiz.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Quiz | Quiz Master</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}"/>
</head>
<body>
  <header class="top-bar">
    <span class="logo-sm">⬚ Quiz Master</span>
    <span id="timer" class="timer">⬚ 00:00</span>
  </header>

  <div class="quiz-container">
    <h2 class="quiz-title">Answer All Questions</h2>

    <form action="/submit" method="POST" id="quizForm">
      {% for i, q in enumerate(questions) %}
      <div class="question-card">
        <div class="q-header">
          <span class="q-number">Q{{ i + 1 }}</span>
          <span class="q-progress">{{ i + 1 }} / {{ total }}</span>
        </div>
        <p class="q-text">{{ q.question }}</p>
        <ul class="options">
          {% for opt in q.options %}
          <li>
            <label class="option-label">
              <input type="radio" name="q{{ i }}" value="{{ opt }}" required />
              <span class="option-text">{{ opt }}</span>
            </label>
          </li>
          {% endfor %}
        </ul>
      </div>
      {% endfor %}

      <div class="submit-row">
        <button type="submit" class="btn-submit">Submit Quiz 🚀</button>
      </div>
    </form>
  </div>

  <script>
    let seconds = 0;
    setInterval(() => {
```

```
      seconds++;
      const m = String(Math.floor(seconds / 60)).padStart(2, "0");
      const s = String(seconds % 60).padStart(2, "0");
      document.getElementById("timer").textContent = `⏱ ${m}:${s}`;
    }, 1000);

    document.querySelectorAll('.option-label input').forEach(radio => {
      radio.addEventListener('change', function () {
        const card = this.closest('.question-card');
        card.querySelectorAll('.option-label').forEach(l =>
l.classList.remove('selected'));
        this.closest('.option-label').classList.add('selected');
      });
    });
  </script>
</body>
</html>
```

Result.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Result | Quiz Master</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}"/>
</head>
<body>
  <div class="result-container">
    <div class="score-card">
      <div class="grade-badge">{{ grade }}</div>
      <h1 class="score-heading">{{ score }} / {{ total }}</h1>
      <p class="score-percent">{{ percentage }}%</p>
      <p class="feedback">{{ feedback }}</p>
      <div class="progress-bar-wrap">
        <div class="progress-bar" style="width: {{ percentage }}%;"></div>
      </div>
    </div>

    <h3 class="review-heading">📋 Review Your Answers</h3>

    {% for r in results %}
    <div class="review-card {{ 'correct-card' if r.is_correct else 'wrong-card'
}}">
      <p class="review-q">{{ loop.index }}. {{ r.question }}</p>
      <p class="review-ans">
        Your answer:
        <strong class="{{ 'green' if r.is_correct else 'red' }}">{{ r.selected
}}</strong>
      </p>
```

```
      {% if not r.is_correct %}
      <p class="review-correct">✅ Correct: <strong>{{ r.correct }}</strong></p>
      {% endif %}
    </div>
    {% endfor %}

    <div class="result-actions">
      <a href="/restart" class="btn-restart">🔄 Try Again</a>
      <a href="/" class="btn-home">🏠 Home</a>
    </div>
  </div>
</body>
</html>
```

Style.css:

```
*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }

body {
  font-family: 'Segoe UI', sans-serif;
  background: #f4f6fb;
  color: #2d2d2d;
  min-height: 100vh;
}

.hero {
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
  padding: 20px;
}

.hero-card {
  background: #ffffff;
  border-radius: 20px;
  padding: 50px 40px;
  text-align: center;
  max-width: 480px;
  width: 100%;
  box-shadow: 0 8px 32px rgba(100, 120, 200, 0.10);
}

.logo { font-size: 4rem; margin-bottom: 12px; }

h1 {
  font-size: 2.2rem;
  font-weight: 700;
  color: #3b5bdb;
  margin-bottom: 10px;
}
```

```
.subtitle {
  color: #6b7280;
  line-height: 1.6;
  margin-bottom: 28px;
  font-size: 0.98rem;
}

.info-row {
  display: flex;
  justify-content: center;
  gap: 14px;
  flex-wrap: wrap;
  margin-bottom: 32px;
}

.info-box {
  background: #eef2ff;
  border-radius: 10px;
  padding: 10px 16px;
  font-size: 0.88rem;
  color: #3b5bdb;
  display: flex;
  align-items: center;
  gap: 6px;
}

.btn-start {
  display: inline-block;
  background: #3b5bdb;
  color: #fff;
  padding: 14px 40px;
  border-radius: 50px;
  text-decoration: none;
  font-size: 1rem;
  font-weight: 600;
  transition: background 0.2s, transform 0.2s;
}
.btn-start:hover { background: #2f4abf; transform: translateY(-2px); }

.top-bar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 14px 24px;
  background: #ffffff;
  border-bottom: 1px solid #e5e7eb;
  position: sticky;
  top: 0;
  z-index: 10;
}
```

```
.logo-sm { font-weight: 700; color: #3b5bdb; font-size: 1.1rem; }
.timer { font-size: 0.95rem; color: #6b7280; font-weight: 500; }

.quiz-container {
  max-width: 700px;
  margin: 30px auto;
  padding: 0 16px 60px;
}

.quiz-title {
  text-align: center;
  color: #3b5bdb;
  font-size: 1.4rem;
  margin-bottom: 24px;
}

.question-card {
  background: #ffffff;
  border-radius: 14px;
  padding: 24px;
  margin-bottom: 20px;
  box-shadow: 0 2px 12px rgba(0,0,0,0.06);
  border-left: 4px solid #3b5bdb;
}

.q-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 12px;
}

.q-number {
  background: #eef2ff;
  color: #3b5bdb;
  padding: 4px 12px;
  border-radius: 20px;
  font-size: 0.82rem;
  font-weight: 600;
}

.q-progress { font-size: 0.82rem; color: #9ca3af; }

.q-text {
  font-size: 1.02rem;
  font-weight: 500;
  color: #1f2937;
  margin-bottom: 16px;
  line-height: 1.5;
}
```

```
.options { list-style: none; display: flex; flex-direction: column; gap: 10px; }

.option-label {
  display: flex;
  align-items: center;
  gap: 12px;
  padding: 12px 16px;
  border: 1.5px solid #e5e7eb;
  border-radius: 10px;
  cursor: pointer;
  transition: border-color 0.2s, background 0.2s;
}
.option-label:hover { border-color: #93c5fd; background: #eff6ff; }
.option-label.selected { border-color: #3b5bdb; background: #eef2ff; }
.option-label input[type="radio"] { accent-color: #3b5bdb; width: 16px; height:
16px; }
.option-text { font-size: 0.95rem; color: #374151; }

.submit-row { text-align: center; margin-top: 10px; }

.btn-submit {
  background: #3b5bdb;
  color: #fff;
  border: none;
  padding: 14px 48px;
  border-radius: 50px;
  font-size: 1rem;
  font-weight: 600;
  cursor: pointer;
  transition: background 0.2s, transform 0.2s;
}
.btn-submit:hover { background: #2f4abf; transform: translateY(-2px); }

.result-container {
  max-width: 680px;
  margin: 30px auto;
  padding: 0 16px 60px;
}

.score-card {
  background: #ffffff;
  border-radius: 20px;
  padding: 40px 30px;
  text-align: center;
  margin-bottom: 30px;
  box-shadow: 0 4px 20px rgba(59, 91, 219, 0.10);
}

.grade-badge {
  display: inline-block;
```

```
  background: #eef2ff;
  color: #3b5bdb;
  font-size: 2rem;
  font-weight: 800;
  width: 72px;
  height: 72px;
  line-height: 72px;
  border-radius: 50%;
  margin-bottom: 16px;
}

.score-heading { font-size: 2.8rem; color: #1f2937; font-weight: 800; }
.score-percent { font-size: 1.1rem; color: #6b7280; margin: 6px 0 12px; }
.feedback { font-size: 1.1rem; color: #374151; margin-bottom: 20px; }

.progress-bar-wrap {
  height: 10px;
  background: #e5e7eb;
  border-radius: 10px;
  overflow: hidden;
}

.progress-bar {
  height: 100%;
  background: linear-gradient(90deg, #3b5bdb, #60a5fa);
  border-radius: 10px;
}

.review-heading {
  font-size: 1.1rem;
  color: #374151;
  margin-bottom: 14px;
  padding-left: 4px;
}

.review-card {
  background: #fff;
  border-radius: 12px;
  padding: 18px 20px;
  margin-bottom: 12px;
  border-left: 4px solid;
}
.correct-card { border-color: #22c55e; }
.wrong-card   { border-color: #f87171; }

.review-q { font-weight: 500; margin-bottom: 8px; color: #1f2937; }
.review-ans { font-size: 0.92rem; color: #4b5563; margin-bottom: 4px; }
.review-correct { font-size: 0.92rem; color: #16a34a; }

.green { color: #16a34a; }
.red   { color: #dc2626; }
```

```
.result-actions {
  display: flex;
  justify-content: center;
  gap: 16px;
  margin-top: 28px;
  flex-wrap: wrap;
}

.btn-restart, .btn-home {
  padding: 12px 30px;
  border-radius: 50px;
  font-size: 0.95rem;
  font-weight: 600;
  text-decoration: none;
  transition: transform 0.2s;
}
.btn-restart { background: #3b5bdb; color: #fff; }
.btn-restart:hover { background: #2f4abf; transform: translateY(-2px); }
.btn-home { background: #f3f4f6; color: #374151; border: 1px solid #e5e7eb; }
.btn-home:hover { background: #e5e7eb; transform: translateY(-2px); }

@media (max-width: 480px) {
  .hero-card { padding: 36px 22px; }
  h1 { font-size: 1.8rem; }
  .score-heading { font-size: 2rem; }
  .question-card { padding: 18px 16px; }
}
```

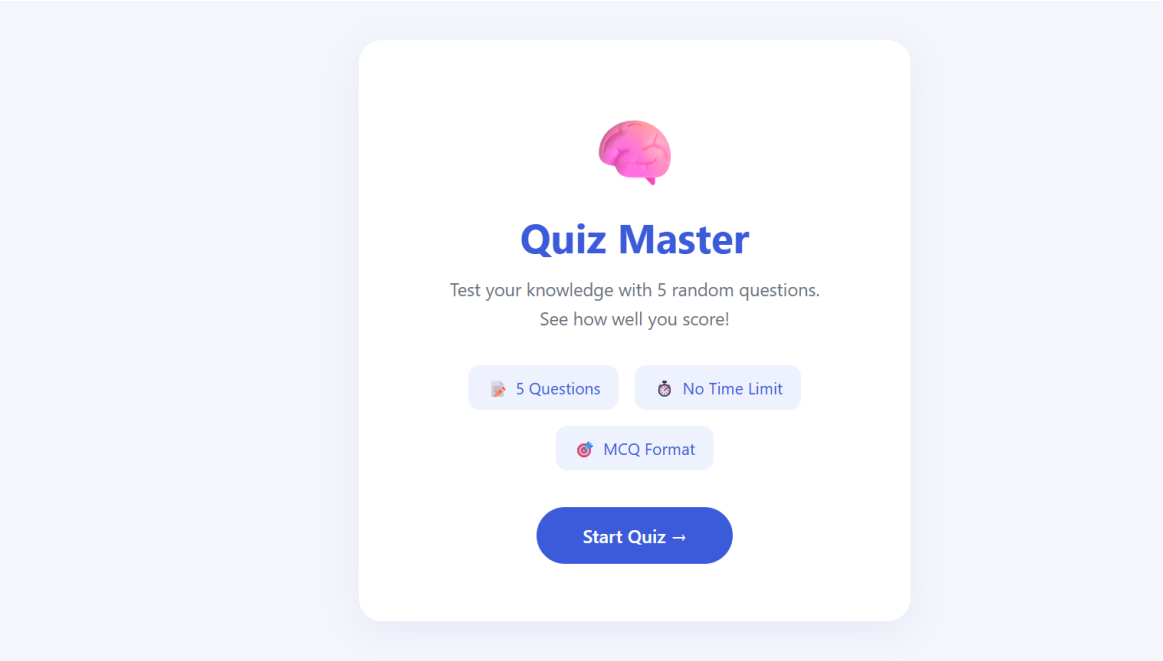
App.py:

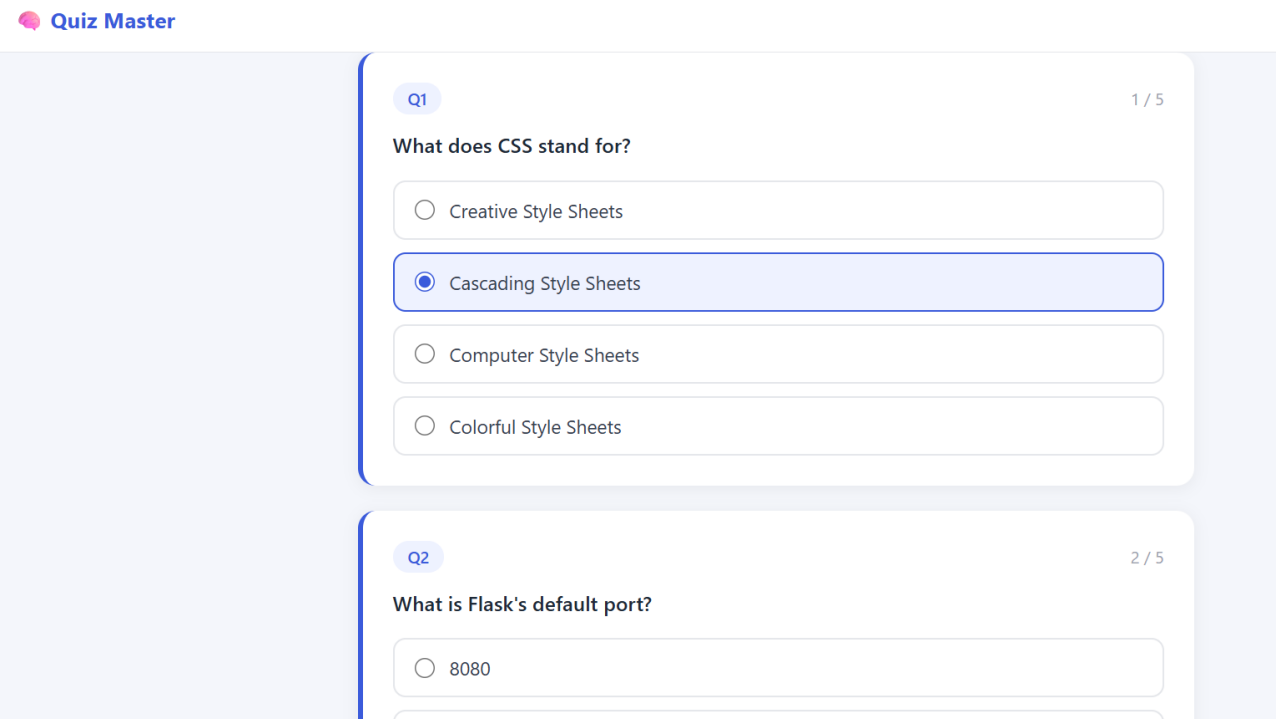
```
from flask import Flask, render_template, request, session, redirect, url_for
import random

app = Flask(__name__)
app.secret_key = "quizmaster_secret"
app.jinja_env.globals['enumerate'] = enumerate

QUESTIONS = [
    {
        "question": "What does HTML stand for?",
        "options": ["HyperText Markup Language", "HighText Machine Language",
                    "Hyperlink and Text Markup Language", "None of the above"],
        "answer": "HyperText Markup Language"
    },
    {
        "question": "Which tag is used to link an external CSS file?",
        "options": ["<style>", "<css>", "<link>", "<script>"],
        "answer": "<link>"
    },
    {
```

```
        "question": "Which Python framework is used to build web apps?",
        "options": ["Django", "Flask", "FastAPI", "All of the above"],
        "answer": "All of the above"
    },
    {
        "question": "What does CSS stand for?",
        "options": ["Creative Style Sheets", "Cascading Style Sheets",
                    "Computer Style Sheets", "Colorful Style Sheets"],
        "answer": "Cascading Style Sheets"
    },
    {
        "question": "Which HTTP method is used to submit a form?",
        "options": ["GET", "POST", "PUT", "DELETE"],
        "answer": "POST"
    },
    {
        "question": "Which of the following is a Python data type?",
        "options": ["int", "float", "list", "All of the above"],
        "answer": "All of the above"
    },
    {
        "question": "What is Flask's default port?",
        "options": ["8080", "3000", "5000", "4200"],
        "answer": "5000"
    },
]

@app.route("/")
def home():
    return render_template("index.html")

@app.route("/quiz")
def quiz():
    questions = random.sample(QUESTIONS, 5)
    session["questions"] = questions
    return render_template("quiz.html", questions=questions, total=len(questions))

@app.route("/submit", methods=["POST"])
def submit():
    questions = session.get("questions", [])
    score = 0
    results = []

    for i, q in enumerate(questions):
        selected = request.form.get(f"q{i}")
        correct = q["answer"]
        is_correct = selected == correct
        if is_correct:
            score += 1
        results.append({
            "question": q["question"],
```

```
            "selected": selected or "Not answered",
            "correct": correct,
            "is_correct": is_correct
        })

    total = len(questions)
    percentage = round((score / total) * 100)

    if percentage == 100:
        feedback = "🏆 Perfect Score! Outstanding!"
        grade = "A+"
    elif percentage >= 80:
        feedback = "🎉 Excellent work! Keep it up!"
        grade = "A"
    elif percentage >= 60:
        feedback = "👍 Good job! Room to improve."
        grade = "B"
    elif percentage >= 40:
        feedback = "📚 Keep practising, you can do it!"
        grade = "C"
    else:
        feedback = "💪 Don't give up! Try again."
        grade = "D"

    return render_template("result.html", score=score, total=total,
                           percentage=percentage, feedback=feedback,
                           grade=grade, results=results)

@app.route("/restart")
def restart():
    session.clear()
    return redirect(url_for("home"))

if __name__ == "__main__":
    app.run(debug=True)
```

output:


Quiz Master
Test your knowledge with 5 random questions.
See how well you score!
5 Questions
No Time Limit
MCQ Format
Start Quiz →


Quiz Master
Q1
1 / 5
What does CSS stand for?
Creative Style Sheets
Cascading Style Sheets
Computer Style Sheets
Colorful Style Sheets
Q2
2 / 5
What is Flask's default port?
8080

A+

5 / 5

100%

🏆 Perfect Score! Outstanding!

📋 Review Your Answers

1. What does CSS stand for?

Your answer: **Cascading Style Sheets**

2. What is Flask's default port?